

Rakshith Raghavendra

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Profile Summary

CSE–AI/ML undergraduate with hands-on experience in AI/ML, Computer Vision, cybersecurity, robotics, and web development. Experienced in building practical AI and software systems, with a focus on solving real-world problems. Founder of TRIP, a technology-driven agricultural supply-chain platform concept.

Education

R V College of Engineering

Bengaluru, India

B.E. Computer Science & Engineering (Artificial Intelligence & Machine Learning) Expected 2028 | CGPA: 9.08/10

Alva's College

Moodabidri, India

Pre-University Education

Skills

Programming: C(Intermediate), Python(Intermediate)

AI/ML: Machine Learning, Computer Vision, CNNs, TensorFlow, Scikit-learn

Web: MERN STACK, Flask, REST APIs

Embedded/IoT: ESP32, Arduino

Tools: Git, GitHub, Google Colab, Jupyter Notebook

Professional: Leadership, Team Collaboration, Problem Solving, Analytical Thinking

Academic Projects

AgroSort AI - AI Robotic Fruit Sorting System

(Python, TensorFlow, OpenCV, Flask, ESP32, IoT)

- Developed an AI-powered fruit-sorting system integrating **CNN-based quality classification** with a 4-DOF ESP32-controlled robotic arm.
- Implemented computer-vision processing using **OpenCV** and integrated the AI inference pipeline with a Flask backend.
- Developed web-based robotic control with **virtual joystick, keyboard controls, and Teach & Repeat functionality** for manual operation.
- Integrated ESP32, servo motors, Wi-Fi/HTTP communication, and Flask APIs to bridge AI predictions with physical robotic actuation.
- Designed an event-driven sorting workflow where the conveyor pauses for image capture and classification before automatically sorting the detected fruit.
- Developed the system architecture combining **AI inference, web dashboard, embedded control, database logging, and robotic actuation**.

Sentinel: AI-Powered Python Vulnerability Detection Engine (Python, Scikit-learn, Flask, TF-IDF, CVEFixes v1.0.7, SAST)

- Built an AI-powered Static Application Security Testing (SAST) system to detect potentially vulnerable Python methods without executing source code.
- Trained and evaluated **Logistic Regression** models using word-level TF-IDF features extracted from real-world vulnerable and patched code from the CVEFixes v1.0.7 dataset.
- Engineered security features combining **NLOC, token count, cyclomatic complexity, nesting depth, and vulnerability-specific heuristics**.
- Integrated handcrafted security guardrails for patterns involving `eval()`, `exec()`, command execution, SQL queries, and dynamic code execution.
- Developed a **Flask-based web application** supporting Python file and GitHub repository scanning, vulnerability classification, confidence scoring, and CWE mapping.
- Implemented data preprocessing, model training, evaluation, and testing workflows using **Pandas, NumPy, Scikit-learn, Pytest, Git, and GitHub**.

AI Computer Vision Object Counter - Logistics | Python, Machine Learning, Computer Vision

CrediFlow - Credit Tracking Application | Web Development, Financial Data Management

Certifications

Unlox Edge Certification - Web Development | Unlox × Outpro India | Feb 2026

Python - Certificate of Completion | MIMO | Apr 2026

AI Workshop -National Space Day 2026 | Bharat Space Education Research Centre (BSERC) | Aug 2026

Extra-Curricular Activities

National Service Scheme (NSS), RVCE | Volunteer | 2025–26

- Participated in community service, environmental sustainability, and cyber-awareness initiatives.
- Collaborated in student-led activities promoting teamwork, civic responsibility, and community engagement.

Author & Screenwriter

- Published **Quantum Shadows** and currently developing an original feature-film screenplay, with experience in storytelling, character development, and narrative design.